Folkhälsomyndigheten

NCDSim – A Simulation Model for the Future Development of Non-Communicable Diseases

V 3.0



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Abbreviations

ATC Anatomical Therapeutic Chemical

CVD Cardiovascular Disease

ICD International Classification of Diseases

IHE Institutet för Hälso- och Sjukvårdsekonomi (Swedish Institute for Health Economics)

PAF Population attributable fraction

RTB Registret över totalbefolkningen (Total Population Register)

SCB Statistiska centralbyrån (Statistics Sweden)

Background

The Public Health Agency of Sweden, in collaboration with the Swedish Cancer Society and the Heart-Lung Foundation, has developed a simulation model, NCDSim, with a web-based user interface. This tool makes it possible to study how cancer and cardiovascular diseases may develop in the future and how various measures targeting the most important risk factors for these diseases could influence this development.

NCDSim is based on Statistics Sweden's (SCB) population projection (Statistics Sweden, 2025) of future population development and can illustrate the potential development of these diseases up to the year 2120. For many questions, however, it is sufficient to make projections over a somewhat shorter time frame.

A starting point for the project is that NCDSim should be based on the best available empirical data regarding population composition, the prevalence of risk factors, and disease cases. The estimates are based on data from SCB, the National Board of Health and Welfare, the Public Health Agency of Sweden, and the Swedish Food Agency. For relative risks of disease incidence based on the presence of risk factors included in the model, relevant results have been sourced from the academic literature.

Purpose

NCDSim is designed to serve as a support tool for exploratory analyses, where assumptions can be relatively easily changed to create different scenarios for future development. Typical questions that can be explored using scenario simulations include how preventive measures targeting one or more of the risk factors included in the model might affect the future development of these diseases. One application of the model is to provide a basis for assessing whether various goals, such as those under the Agenda 2030 framework, are achievable and what measures would need to be implemented to achieve them.

NCDSim provides annual results on the number of ongoing disease cases, the number of new cases, the number of deaths, and healthcare costs for the diseases in question. The results are presented in a user-friendly manner in the model's web interface but can also be exported for further analysis.

Method

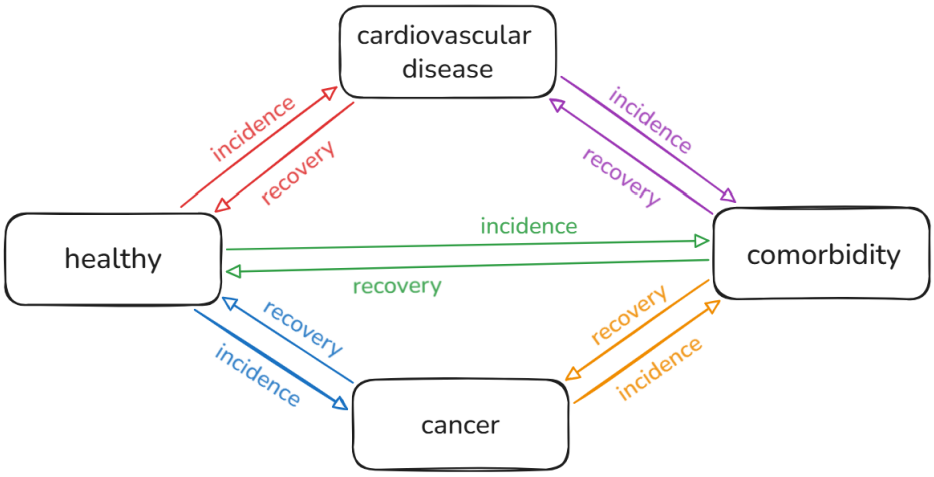
NCDSim's structure and content

NCDSim is a system dynamics model, which means the model handles stocks and flows. Here, stocks refer to the number of individuals who share common characteristics at a given time, such as having a particular disease. Stocks can also be called compartments or states. Flows refer to the number of individuals moving between two stocks, or changing states, over a time interval. Examples of flows in NCDSim include the number of individuals who fall ill with a particular disease or recover from a particular disease during a year.

While stocks represent count data at a specific point in time, flows represent a rate of change or intensity over a time interval. In NCDSim, the model is simulated in time steps corresponding to calendar years. Stocks refer to the number of individuals in each state as of December 31 of each year, while flows refer to the number of people moving between two stocks during the same calendar year.

In NCDSim, individuals are not represented individually but rather as groups of individuals. All individuals included in a stock have exactly the same characteristics, and the only relevant information is thus the number of individuals included in the stock. Similarly, the flows between stocks are also completely homogeneous.

Figure 1 - Stock and flow diagram for NCDSim



NCDSim has four stocks (the boxes in the figure):

1. Cancer,
2. Cardiovascular disease (CVD),
3. Comorbidity (both cancer and CVD),
4. Healthy (individuals without cancer or CVD).

It is important to note that individuals in the latter stock may suffer from other diseases besides cancer or cardiovascular disease, even though they are referred to as "healthy" in the following.

The ten flows (the arrows in the figure) represent the epidemiological events of incidence and recovery. In addition to these epidemiological events, NCDSim also simulates a number of demographic events: births, deaths, immigration, and emigration. Deaths are assumed to occur from all stocks, migration to and from all stocks, while births are assumed to occur only into the healthy stock.

An important aspect of NCDSim is the ability to simulate future demographic development in a realistic way. The increasing proportion of elderly in the population, according to SCB's forecast, is expected to have a major impact on the disease burden. Because of this, NCDSim has been designed with separate modeling according to the stock and flow diagram above. Age is represented in NCDSim in one-year age classes from 0 to 99, and then in an open-ended age group of 100+. This means the model contains 202 sets of stocks and flows. This can be seen as each stock and flow containing 202 cells for storing sex- and age-specific count data. Most assumptions regarding, for example, risk factor prevalence are also made per sex and age.

When the model has simulated the changes in stocks and corresponding flows that occur during a calendar year, the population frequency in each of the 202 cells is moved to the cell representing one year older, within each stock. However, in the cell for individuals aged 100 and older, the frequencies for surviving individuals accumulate.

The model's flows are mathematically described as the product of a risk population and an intensity rate. The risk population more formally refers to the risk time generated in a population, such as the number of person-years. The intensity rate refers to the number of events per unit of time, such as the number of deaths per person-year. Intensity rates are estimated as the number of events, for example, of a demographic nature, divided by the risk time generated in a particular sex and age category. For the demographic flows, good agreement with SCB's population projection is achieved by calculating intensity rates from the projected population and its demographic components of change.

For NCDSim to simulate a realistic development of disease trends, it is important to account for differences in mortality by sex and age between stocks with and without disease. Separate death risks have therefore been estimated from SCB and National Board of Health and Welfare data for each stock and are used in the simulations.

The flows representing incidence are instead calculated by multiplying the risk time within each sex and age category among the non-diseased population by a number proportional to the total population attributable fraction for incidence of each non-communicable disease. The method is described in more detail below.

The flows representing recovery are based on estimates of average annual intensity rates for recovery, as described below. Separate estimates are made for cancer, cardiovascular disease, and comorbidity, as well as by sex and age group.

NCDSim has been implemented in the R programming language, and the graphical user interface in the R package Shiny.

Data sources and definitions

The definitions of cancer and cardiovascular disease used in NCDSim largely follow the definitions used by Freisling et al. (2020), which were used as the primary source of information during the initial phase of model development. Both disease categories encompass a relatively large number of individual diagnoses and are therefore, to some extent, heterogeneous from a clinical perspective and thus in terms of incidence, prevalence, disease duration, and mortality. NCDSim simulates the expected development for the diagnosis groups as a whole and not for individual diagnoses within the group. An assumption is that the composition of diagnoses within the groups is constant over time.

The estimates of prevalence, incidence, and death risks used as input data in NCDSim have been made using individual-level data from SCB (Total Population Register) and the National Board of Health and Welfare (Patient Register for inpatient and specialized outpatient care, as well as the Cancer Register). In these calculations, NCDSim's stocks, i.e., the number of prevalent disease cases for each historical year, have been defined as follows:

1. Cancer patient stock: The number of individuals with at least one registered malignant primary tumor (with ICD-10 codes C00-C97, D00-D09) in the Cancer Register during the year or during any of the four preceding years, and who do not meet the diagnosis code criteria according to point 2 below.
2. Cardiovascular disease patient stock: The number of individuals with at least one healthcare visit registered in the Patient Register with a main or secondary diagnosis for any of the ICD-10 codes B33, F01, I01-I09, I20-I26, I27, or I28-I73 during the year or during any of the four preceding years, and who do not meet the diagnosis code criteria according to point 1 above.
3. Comorbidity stock: The number of individuals who simultaneously meet the diagnosis code criteria according to points 1 and 2 above.
4. Healthy stock: The difference between the total number of individuals in the population, according to SCB's Total Population Register, and the sum of stocks 1, 2, and 3.

Note that the stocks are defined based on a five-year period prevalence, where the method differs between diseases due to different data sources being used. The method for defining an incident case also differs for the same reason.

For cancer, information on newly registered malignant tumors from the Cancer Register is used. An individual who has at least one registered malignant primary tumor during a year is defined as an incident case. An individual is defined as a

prevalent case during the incidence year and the following four years. If a new malignant primary tumor is registered during the prevalence period, which is relatively rare, the period restarts without the individual being considered an incident case again.

For cardiovascular disease, data on healthcare visits and diagnosis codes from the Patient Register are used instead. An individual is defined as a prevalent case if the individual has had at least one healthcare visit with a relevant ICD code during the current year or any of the four preceding years. If an individual is not a prevalent case, they are at risk for incidence, which means the individual is counted as an incident case if there is at least one healthcare visit during the year. Note that the individual is simultaneously defined as a prevalent case.

It can be noted that, based on the definitions above, individuals may have multiple periods of uninterrupted five-year prevalence, or time in disease, lasting more than five years. For example, an individual who has a healthcare visit every fourth year over a longer period will be counted as an incident case only at the first visit and thereafter as a prevalent case up to and including four years after the last visit, regardless of how much time has passed since the individual was an incident case. Longer periods of time in disease are significantly more common in cardiovascular disease than in cancer.

Recovery intensity rates are empirically estimated by dividing the number of individuals who recover during a year by the number of person-years lived in the relevant risk population. Recovery refers to an individual moving from being prevalent in a particular disease at the end of one year to being in the healthy stock at the end of the next year, or, for individuals with comorbidity at the end of one year, transitioning to one of the stocks with only one disease by the end of the next year. A total of five intensity rates for recovery are estimated, by age and sex, corresponding to the following state transitions:

- From comorbidity to cancer,
- From comorbidity to cardiovascular disease,
- From comorbidity to healthy,
- From cancer to healthy,
- From cardiovascular disease to healthy.

Definition and Prevalence of Risk Factors

The prevalence of the various risk factors included in NCDSim has been estimated from two data sources:

- The Public Health Agency of Sweden's National Public Health Survey, *"Hälsa på lika villkor?"* (Public Health Agency of Sweden, 2024), for non-dietary risk factors,

- The Swedish Food Agency's Riksmaten Survey (Swedish Food Agency, 2011), for dietary risk factors.

The estimates have, as far as possible, taken into account differences between sexes and ages. Risk groups are not represented by explicit stocks in NCDSim but are calculated as proportions of the population that do not have cancer or cardiovascular disease and are thus at risk for disease incidence. The risk factors are considered independent, and the calculation of the size of risk groups can therefore be done separately for each risk factor. Based on each risk factor, the population without disease is divided into two groups:

- A risk group with a relative risk greater than one,
- A reference group with a relative risk equal to one.

Thresholds for dichotomization are shown in Table 1.

For dietary risk factors, thresholds have been chosen, as far as possible, to align with the Nordic Nutrition Recommendations. For non-dietary risk factors, thresholds for dichotomization have been chosen, where possible, to align with the definition in the study where the relative risk was estimated. In cases where it has not been possible to use the same definitions, the relative risk has been adjusted so that it can be used for the dichotomization applied in NCDSim.

Table 1 – Average prevalence of risk factors (proportions) among individuals aged 16–85 years.

Risk Factor	Men	Women	Source	Threshold
Low fruit consumption	0.83	0.71	Riksmaten (Adults)	< 200 g/day
Low whole grain consumption	0.89	0.94	Riksmaten (Adults)	< 100 g/day
Low vegetable consumption	0.84	0.78	Riksmaten (Adults)	< 200 g/day
High consumption of red and processed meat	0.81	0.66	Riksmaten (Adults)	> 50 g/day
High salt consumption	0.85	0.62	Riksmaten (Adults)	> 6 g/day
High BMI	0.55	0.43	HLV	BMI > 25
Smoking[1]	0.11	0.10	HLV	Smoker[1]
Insufficient physical activity	0.31	0.30	HLV	< 150 min/week of moderate-intensity activity
Alcohol consumption	0.20	0.10	HLV	> 12 g/day

Ultraviolet radiation (UV radiation) differs from the other risk factors in that we do not start from an empirically estimated risk factor prevalence and a relative risk.

[1] Smokers include both daily and occasional smokers.

Instead, its effect in the model is based directly on an attributable fraction drawn from the literature.

Relative Risks for Disease Incidence

In NCDSim, relative risks are used for the presence of risk factors to simulate how many cases of disease are generated, i.e., disease incidence. The methodology is described in more detail in the next section. Relative risks have been sourced from articles published in scientific journals.

Table 2 presents the risk factors included in the model, along with information on the specific diagnosis codes within each disease category that form the basis for risk estimation, as well as the studies used as sources.

Table 2 - Risk Factors, Specific Diagnosis Codes (ICD-10), Literature References, and Relative Risks for Cancer and Cardiovascular Disease (CVD)

Risk Factor	Disease (ICD-10 Codes)	Source	RR (Cancer)	RR (CVD)
Low fruit consumption	Cancer (C15, C33-C34), CVD (I20-I25, I60-I63)	GBD 2019 (2020)	1.004	1.029
Low vegetable consumption	Cancer (C15), CVD (I20-I25, I60-I63)	GBD 2019 (2020)	1.000	1.024
Low whole grain consumption	Cancer (C15, C33-C34), CVD (I20-I25, I60)	GBD 2019 (2020)	1.012	1.063
High consumption of red and processed meat	Cancer (C50, C18-C20), CVD (I20-I25, I60-I63)	GBD 2019 (2020)	1.021	1.078
High salt consumption	CVD (B33, F01, I01-I09, I20-I26, I27, I28-I73)	Wang et al. (2020)	1.000	1.150
Smoking	Cancer (C00-D09)	Mons et al. (2015)	1.340	-
Smoking	CVD (B33, F01, I01-I09, I20-I26, I27, I28-I73)	SCORE2 Working Group (2021)	-	1.850
Insufficient physical activity	Cancer (C00-D09), CVD (F01, I01-I09, I20-I79)	Garcia et al. (2023)	1.090	1.250
High BMI (≥25)	Cancer (C00-D09), CVD (B33, F01, I01-I09, I20-I26, I27, I28-I73)	Freisling et al. (2020)	1.040	1.270
Alcohol consumption	Cancer (C00-C15, C18-C20, C22, C32-C33), CVD (I48)	Carr et al. (2024); Brauer et al. (2024); Lankester et al. (2021)	1.042	1.046
UV radiation*	Cancer (C43, C44)	Langselius et al. (2025)	0.077	-

* For UV radiation the population attributable fraction (PAF), instead of a relative risk, is reported.

For some risk factors, relative risks are not reported collectively for all diagnoses included in the cancer and cardiovascular disease categories as defined in NCDSim, but only for a subset of diagnosis codes. This applies to several of the

dietary risk factors and to insufficient physical activity. In these cases, the relative risks have been weighted against the proportion of the total disease prevalence within each disease category that is accounted for by the relevant diagnosis codes.

For many of the risk factors, the relative risks in the studies have been estimated using a different categorization than that used in NCDSim. In some cases, there have been more than two categories, or a dichotomization at a different threshold. In some cases, hazard ratios have also been estimated with the risk factor as a continuous variable. In all cases, a recalculation of the estimated relative risks has been carried out so that the relative risks can be applied to NCDSim's dichotomous risk groups.

Example: Estimating Relative Risk and Prevalence for Alcohol

Data Collection and Unit Conversion

Data on alcohol consumption is sourced from two questions in the Public Health Agency of Sweden's National Public Health Survey (HLV). The responses are converted to grams of alcohol per day using estimated values for frequency and quantity per occasion, as shown in the tables below[2]:

Question 37: How often have you consumed alcohol in the past 12 months?		
Response Option	Times per week	Times per day
4 times/week or more	5	0.71
2–3 times per week	2.5	0.36
2–4 times per month	0.67	0.10
1 time/month or less	0.1	0.01
Never	0	0

Question 38: How many "drinks" did you typically consume on a day when you drank alcohol in the past 12 months?	
Response option	**Drinks per occasion**
1–2	1.5
3–4	3.5
5–6	5.5
7–9	8

[2] The questions and answers have been translated to English from the original Swedish version in the survey for clarity.

Question 38: How many "drinks" did you typically consume on a day when you drank alcohol in the past 12 months?	
10 or more	11

Calculating daily alcohol consumption

For each individual, daily alcohol consumption is calculated by multiplying frequency (times/day), quantity (drinks/occasion) and standard measure of 12 g/drink. This provides an estimate of grams of alcohol per day (g/day).

Definition of Risk Group and Reference Group

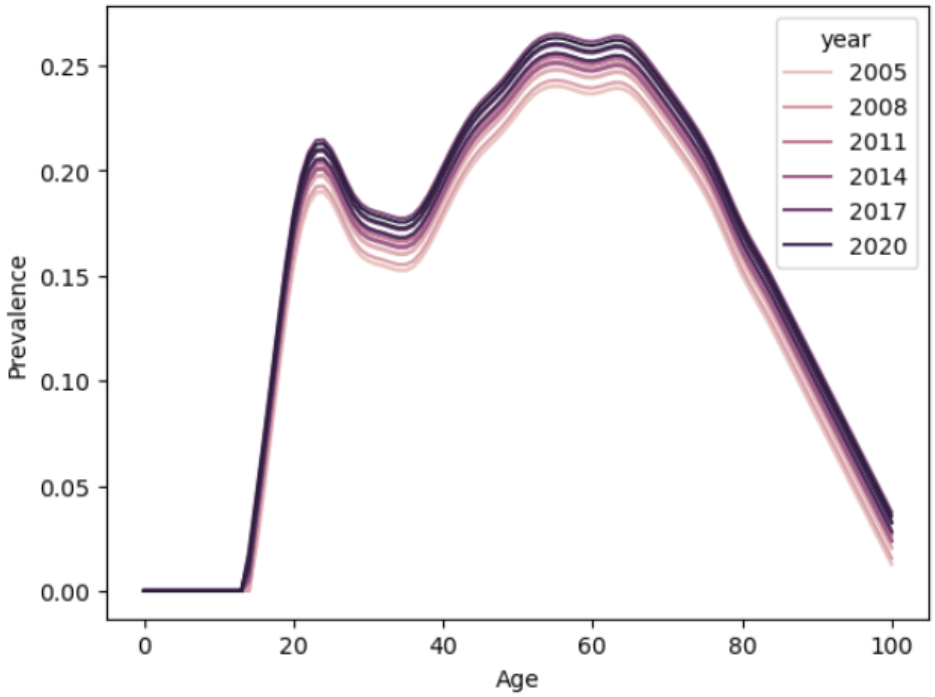
In NCDSim, the effect of risk factors on the incidence of cancer and cardiovascular disease is implemented by dividing the population into a risk group and a reference group, i.e., a dichotomous division. For alcohol, a consumption exceeding the threshold of 12 g/day, i.e., one standard drink, has been used to classify the risk group.

Estimating Risk Factor Prevalence by Year, Age, and Sex

Based on the estimated daily alcohol consumption, the proportion of individuals, by age and sex, who belong to the risk group (i.e., those with alcohol consumption exceeding 12 g/day) is calculated. The calculation takes into account HLV's sampling and calibration weights, which provide an estimate of the population's risk factor prevalence for alcohol consumption.

Calculated year- and sex-specific age profiles for the proportion of individuals in the risk group show a relatively large variation between adjacent ages. To smooth out the age profiles and obtain better estimates, generalized additive models (GAMs) have been used. This method allows the age profile to be modeled in a highly flexible manner (using splines) while accounting for how alcohol consumption differs between sexes and over time. The shape of the age profile is assumed to be the same for all years but is allowed to differ in level between sexes and over time.

Figure 2: Proportion of men drinking more than 12 g/day of alcohol, by age and year. Both estimated and extrapolated from HLV data



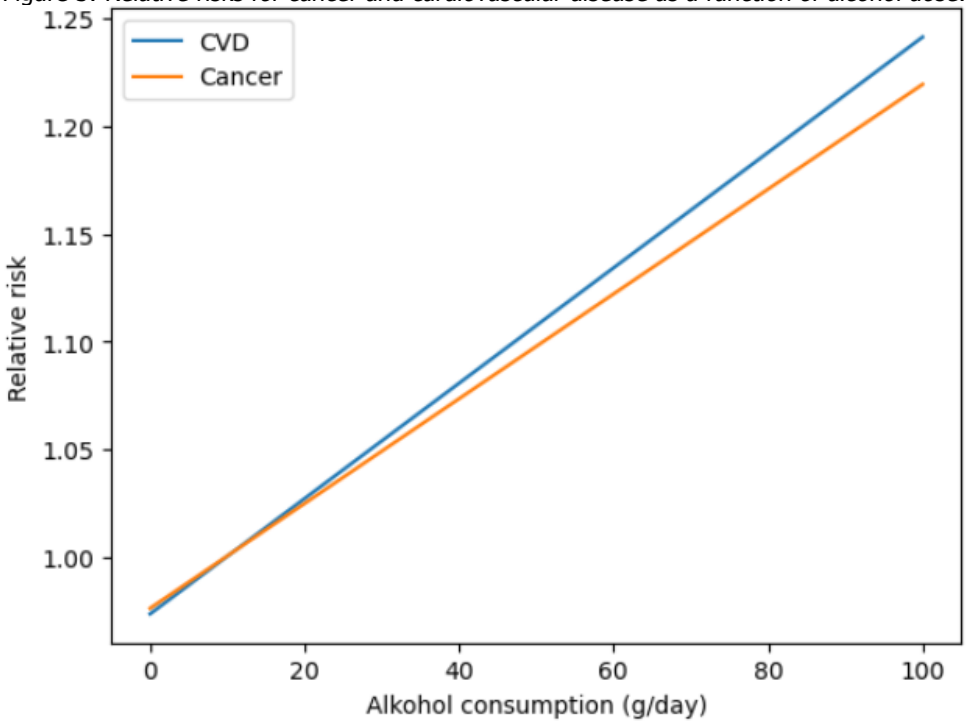
Calculating Relative Risks

The relative risks for incidence of cancer and cardiovascular disease based on alcohol consumption levels used in NCDSim have been sourced from the Global Burden of Disease (GBD) study and processed to adapt them to NCDSim's dichotomous division into risk group vs. reference group.

The relative risks refer to specific diagnosis codes and first need to be adapted to the broader disease groups of cancer and cardiovascular disease modeled in NCDSim. This adaptation is done by weighting the relative risk for each diagnosis code based on the diagnosis code's relative share of the total prevalence of the disease group. Data on the prevalence of different diagnosis codes have been sourced from the National Board of Health and Welfare's Patient Register (for cardiovascular disease) and the Cancer Register (for cancer). Note that this method implies an implicit assumption that the relative risks for diagnosis codes included in each disease group but for which no relative risks are estimated in GBD are equal to one.

The relative risks reported in GBD have been estimated for nine groups with respect to alcohol consumption: 0–96 grams per day, in increments of 12 grams. In a first step, a linear dose-response curve is fitted to the average consumption within each group (assumption) and the groups' estimated relative risks. In the figure below, the relative risk for incidence is shown as a linear function of increased alcohol consumption in grams per day, where zero grams of alcohol consumption is the reference category (the relative risk is equal to one).

Figure 3: *Relative risks for cancer and cardiovascular disease as a function of alcohol dose.*



To estimate the dichotomous relative risk, the following steps are taken:

The average consumption in the risk group (i.e., the proportion of the population drinking more than 12 g/day of alcohol, x_1) is estimated, and the relative risk for the risk group, i.e., $RR(x_1)$, is estimated based on the estimated relationship between alcohol intake and relative risk.

The step is repeated for the reference group (x_0), and $RR(x_0)$ is calculated.

The dichotomous relative risk for those consuming more than 12 g of alcohol per day is then $RR(x_1) / RR(x_0)$. This involves a normalization of the relative risks so that the relative risk for the reference group is equal to one.

Once the prevalence of the risk factor (proportion drinking more than 12 g/day) and the corresponding relative risks have been calculated, they can be used as input in NCDSim to simulate the impact of alcohol consumption on public health. This makes it possible to estimate the total disease burden attributable to alcohol consumption.

After the above-described adjustments to the estimated relative risks, relative risks for each risk factor and disease group are obtained that can be used to simulate disease incidence according to the method described in the next section.

Simulating Incidence Based on Attributable Fractions

The total contribution to the risk of disease incidence that can be attributed to a risk factor depends on both the prevalence of the risk factor and the increased risk given the presence of the risk factor. In NCDSim, the concept of population attributable

fraction (PAF) (see, e.g., Bouter et al., 2018) is used as a way to quantify the incidence contribution of risk factors. For each risk factor, a PAF is calculated that refers to the risk factor's contribution to the total disease incidence.

Populations with the presence of risk factors are not represented by explicit stocks in the model but are calculated as a proportion of the stock without disease. In the following, more formal description, J denotes the number of risk factors with prevalence $p^{j}_{s,a}$ for risk factor j, sex s, and age a. For each risk factor, the stock of individuals without disease can be divided into two groups:

- Individuals with increased risk of disease incidence,
- A reference group.

For cancer, cardiovascular disease, and comorbidity, the relative risk for each risk factor is assumed to be independent of sex and age and is denoted by RR^{j}. The contribution to the total incidence, PAF, for risk factor j, sex s, and age a is calculated as:

$$PAF^{j}_{s,a} = \frac{p^{j}_{s,a}\left(RR^{j} - 1\right)}{1 + p^{j}_{s,a}(RR^{j} - 1)}$$

The total incidence contribution, or combined PAF, from the risk factors included in NCDSim for sex s and age a, denoted $CPAF_{s,a}$, is calculated as:

$$CPAF_{s,a} = 1 - \left(1 - w_1 PAF^{1}_{s,a}\right)\left(1 - w_2 PAF^{2}_{s,a}\right) \ldots \left(1 - w_J PAF^{J}_{s,a}\right)$$

where $w_j = 1 - C_j$. The so-called communality C_j for the j-th risk factor refers to the degree of overlap or covariation with other risk factors. The higher the degree of overlap, the more $PAF^{j}_{s,a}$ needs to be reduced to avoid an overestimation of $CPAF_{s,a}$. Estimating Cj requires data on all risk factor prevalences $p^{j}_{s,a}$ *(see, e.g., Norton et al., 2014).* Since not all risk factors have been observed in the same data material, reliable estimates of C_j have not been possible, and in the model, in its current version, independence between risk factors is assumed, i.e., $w_j = 1$.

It can be noted that the definition of $CPAF_{s,a}$, means that the aggregation of PAFs is done multiplicatively, in a way that ensures that the combined PAF is always a number less than one. This simultaneously means that the total PAF cannot be expressed as a sum of the individual risk factors' contributions.

PAFs and combined PAFs are calculated for each combination of sex and age, separately for cancer, cardiovascular disease, and comorbidity, and are used to simulate disease incidence. When calculating PAFs for incidence of comorbidity from the cancer and cardiovascular disease stocks, the same relative risks are used as for incidence of the relevant disease from the healthy stock. The PAF for, for example, obesity in cancer is calculated using the same relative risk for incidence of cancer from the healthy stock as from the cardiovascular disease stock (i.e., for inflow to the comorbidity stock). In the current version of the model, risk factor prevalence is estimated generally in the population, by sex and age. This means

that not only are the relative risks for incidence of cancer from the healthy stock and from the cardiovascular disease stock the same, but so are the PAFs. However, development work is underway to introduce, in a future version of the model, estimates that account for differences in the prevalence of risk factors between the different stocks.

As the combined PAF for incidence of comorbidity directly from the healthy stock, the largest of the combined PAFs for incidence of cancer and cardiovascular disease, respectively, is used by sex and age.

The combined attributable fractions described above are calibrated to ensure that the model reproduces the historically observed incidence, by adjusting the magnitude of a residual attributable fraction representing factors that cannot be attributed to the risk factors in the model, such as chance, genetic factors, and other risk factors. The calibration is performed by sex and age, separately for each disease stock. The calibration constants are denoted $PAF^0_{a,s}$ in the equation below, where *a* denotes age and *s* denotes sex.

Under NCDSim's default settings, the residual attributable fraction accounts for up to 81 per cent of cancer cases and 56 per cent of cardiovascular cases, which means that 19 per cent and 44 per cent of disease cases can be attributed to the model's risk factors, for cancer and cardiovascular disease respectively[3].

There is also an option to calibrate the combined attributable fractions to a predetermined value. The calibration constants are denoted $k_{a,s}$ in the equation and ensure that the number of new cases, for each disease, is proportional to the attributable fraction for the given risk factor. The calculated intensity rate used to simulate incidence in the model for sex *s* and age *a*, $P_{a,s}$, can be written as:

$$P_{a,s} = 1 - \left(1 - k_{a,s} * cPAF_{a,s}\right)\left(1 - PAF^0_{a,s}\right)$$

where

$$k_{a,s} = \frac{TF}{\overline{cPAF_{a,s}}} IR_{a,s}$$

and

$$PAF^0_{a,s} = \frac{IR_{a,s} - k_{a,s} * cPAF_{a,s}}{1 - k_{a,s} * cPAF_{a,s}}$$

[3] We obtain this result for the attributable fraction under the assumption that the risk factors affect the risk of developing the disease between the ages of 40 and 80. When the risk factors are applied to the entire age range of the model population, between 0 and 100 years of age, the attributable fractions for cancer and cardiovascular disease become 17 and 37 per cent respectively.

$IR_{a,s}$ is the empirically observed intensity rate for incidence for sex *s* and age *a*. If no calibration to exogenously given PAFs (TF) is performed, the simplified equation is instead obtained:

$$k_{a,s} = IR_{a,s}$$

The calculated calibration coefficients $PAF_{a,s}^{0}$ and $k_{a,s}$ are assumed to be constant over time and between scenarios.

An important aspect of the model development work is validation. The calibration of the combined PAFs is complemented by a check that the model can also generate a disease prevalence that matches the historically observed prevalence.

The underlying mathematical model

The graphical description of the relationship between NCDSim's stocks and flows, as presented in Figure 1, Also has a mathematical counterpart that is used when the model is simulated. For each of the model's stocks $S_{stock}(t)$ a differential equation specifies how the various flow components $F^{flow}_{stock}(t)$ affect the stock at time t. The stocks are denoted below as *healthy*, *cancer*, *cvd* and *com* (comorbidity).

See the equations below:

$$\begin{aligned}\frac{dS_{healthy}(t)}{dt} = {} & F^{born}_{healthy}(t) - F^{dead}_{healthy}(t) + F^{immig}_{healthy}(t) - F^{emig}_{healthy}(t) \\ & + F^{healthy}_{cvd}(t) - F^{cvd}_{healthy}(t) + F^{healthy}_{cancer}(t) - F^{cancer}_{healthy}(t) \\ & + F^{healthy}_{com}(t) - F^{com}_{healthy}(t)\end{aligned}$$

$$\begin{aligned}\frac{dS_{cancer}(t)}{dt} = {} & -F^{dead}_{cancer}(t) + F^{immig}_{cancer}(t) - F^{emig}_{cancer}(t) + F^{cancer}_{healthy}(t) \\ & - F^{healthy}_{cancer}(t) + F^{cancer}_{com}(t) - F^{com}_{cancer}(t)\end{aligned}$$

$$\begin{aligned}\frac{dS_{cvd}(t)}{dt} = {} & -F^{dead}_{cvd}(t) + F^{immig}_{cvd}(t) - F^{emig}_{cvd}(t) + F^{cvd}_{healthy}(t) - F^{healthy}_{cvd}(t) \\ & + F^{cvd}_{com}(t) - F^{com}_{cvd}(t)\end{aligned}$$

$$\begin{aligned}\frac{dS_{com}(t)}{dt} = {} & -F^{dead}_{com}(t) + F^{immig}_{com}(t) - F^{emig}_{com}(t) + F^{com}_{healthy}(t) - F^{healthy}_{com}(t) \\ & - F^{cvd}_{com}(t) + F^{com}_{cvd}(t) - F^{cancer}_{com}(t) + F^{com}_{cancer}(t)\end{aligned}$$

This system of equations is solved numerically using the R package deSolve.

It should be noted that the model consists of 202 equation systems like the one above, one for each combination of sex and age group. Within each such combination, solving the equations provides the size of the stocks at the end of the current year. The method also makes it possible to calculate the model's flows during the current year. The process is repeated for each year within the chosen simulation horizon.

Future scenarios

NCDSim has been constructed to enable exploratory analyses where the effects of changed assumptions, for example regarding the prevalence of risk factors in the population, can be easily highlighted. The analyses start from a baseline scenario, which is constructed to serve as the baseline against which the effects of changes can be compared and evaluated. By creating alternative scenarios, with partially different assumptions than in the baseline scenario, the effect of the alternative assumptions can be quantified.

Assumptions in the Baseline Scenario

A fundamental assumption in NCDSim is that the demographic development in the baseline scenario follows the projections made by SCB (Statistics Sweden, 2025). Since the model takes into account that death risks differ between stocks, based on empirically estimated death risks, a calibration of mortality by year, sex, and age must be carried out. The calibration ensures that the average mortality in the population does not, in the long term, deviate from the assumed mortality in SCB's projection, which gradually decreases over time. The calibration preserves the relative mortality differences between stocks.

Furthermore, it is assumed that the empirically observed prevalence of risk factors in the population remains at current levels in the future, per sex and age. If, in a particular analysis situation, one wants to make a trend projection for the prevalence of one or more risk factors, this can be relatively easily implemented through the model's user interface.

The relative risks used to simulate each risk factor's contribution to the total disease incidence are assumed to be unchanged in the future.

The estimated intensity rates for recovery from each disease state are assumed to be unchanged over time, per sex and age.

Design and Analysis of Alternative Scenarios

Through alternative scenarios, the hypothetical effects of interventions can be studied and quantified. Typically, this is done by reducing the prevalence of one or more risk factors as a result of an assumed or estimated effect of a specific intervention. Another application can be to illustrate the negative effects of a continued increase in a risk factor that shows an increasing trend.

In an alternative scenario with a different development of disease compared to the baseline scenario, mortality will also develop differently due to the mortality differences between the healthy and diseased populations. Some effects may also occur on other demographic flows, as these are simulated through intensity rates applied to the population. These are normally second-order effects, as differences in mortality primarily occur at older ages where the impact on migration and nativity is small.

In NCDSim, explicit assumptions about the potential time delay between an intervention and its effects on health outcomes can be made by implementing a (linear) phase-in of the effect on the prevalence of a risk factor over a predetermined number of years.

When analyzing an alternative scenario, it is usually the differences compared to the corresponding baseline scenario that are of primary interest, regardless of which outcome is analyzed – incidence, prevalence, or costs. Whether absolute or relative differences are analyzed often depends on the context and the outcome variable.

Calculating Direct and Indirect Societal Costs

Cost Assumptions

Cost estimates include direct medical costs as well as indirect costs based on production loss during sick leave. All data on resource utilization within healthcare and cost data are from 2024. Future projections in the modeling are presented based on 2024 cost levels, and can be seen as a budget impact analysis; thus, future costs are not discounted. Historically, price development within healthcare has been greater than the overall inflation rate in society, but this has not been adjusted for in the model because it is difficult to predict future relative price developments and to ensure that the analyses more clearly show the cost changes directly linked to changes in incidence and population development. Cost calculations for cancer and cardiovascular disease patients are based on healthcare episodes with a main diagnosis within each group. This means that the stock for comorbidity corresponds to the sum of cancer and cardiovascular disease patients with comorbidity.

Direct costs

Healthcare costs for each diagnosis group are based on data from the KPP (Cost per Patient) database at the Association of Swedish Local Authorities and Regions (SALAR, 2025). The database reports costs per healthcare episode for:

- Specialized somatic care (divided into inpatient care, specialized outpatient care, and primary care),
- Specialized psychiatry.

Only costs where diagnoses are listed as the main diagnosis are included.

The KPP database does not cover all of Sweden's regions. We have made the assumption that the regions included are representative of the country as a whole and have scaled the costs to a national total cost based on the coverage rate of each healthcare level in the database.

Medication costs for prescription drugs are based on the National Board of Health and Welfare's prognosis report (Socialstyrelsen, 2025) for the ATC group cardiovascular drugs (C). The total cost for cancer drugs (L01) has been based on the Swedish Dental and Pharmaceutical Benefits Agency's (TLV) estimates of the

total cost after rebates (agreements between pharmaceutical companies and individual regions) (TLV, 2025). For other drug groups, the rebate is assumed to be negligible, except for PCSK9 inhibitors (within the cardiovascular disease group), whose costs have also been adjusted for rebates.

Indirect costs

Production loss due to disease is based on the Swedish Social Insurance Agency's (Försäkringskassan) public database on the number of net days with paid sick leave and rehabilitation benefits by diagnosis (Försäkringskassan, 2025). The statistics include only sick leave episodes longer than 14 days, as shorter absences are covered by the employer. The age distribution within each diagnosis group is based on data on the number of disease cases per age group. Production loss is estimated by multiplying:

- The number of net days,
- The average daily wage including employer contributions for 2024 for each age group (the average for all age groups is 2,603 SEK per day),
- Adjusted for employment rate, using data from Statistics Sweden's (SCB) public databases (Statistics Sweden, 2025).

Graphical User interface

As mentioned above, the model is written in the R programming language. During model development and for certain analyses, work typically involves writing and executing commands directly in the development environment (usually RStudio) for R. To make the model more practically usable, we have developed a graphical user interface with two main functions:

1. Running new customized simulation runs,
2. Visualizing results from the simulations.

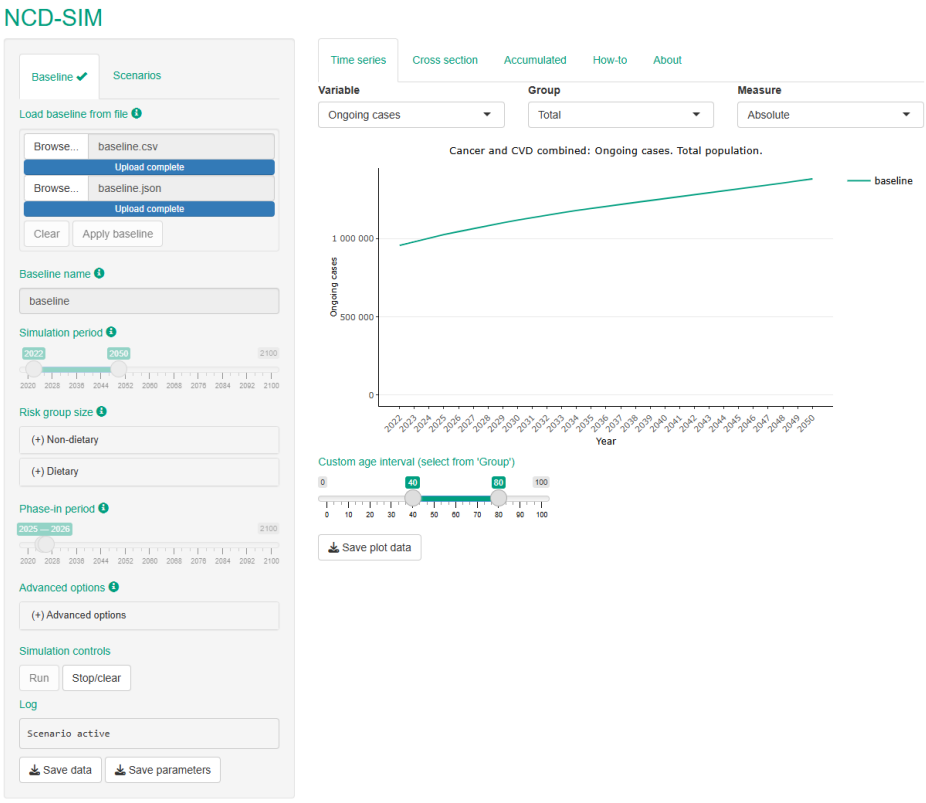
The interface is written in Shiny and functions as a web application with elements of R code. In the current version, the web app is designed to be used entirely on the local computer, via RStudio. However, it is possible to construct a solution where the client computer, via the web app, can run simulations on a server instead of on the local computer. The interface is available in Swedish and English.

The interface consists of a control panel (on the left) and a main panel for displaying results in graph form (see Figure 4 below). Through the control panel, you can make various settings for the simulations, such as:

- Simulation period,
- Risk group size (proportion at risk).

You can then start one or more simulations by having the interface call the simulation script in the background. Once a simulation is complete, the result can be saved to a file to be loaded into the interface for analysis at a later time.

Figure 4 – NCDSim's graphical user interface.



You must always start by simulating a baseline scenario or loading a pre-simulated baseline scenario. All alternative scenarios generated during the same user session use the calibration constants from the loaded baseline scenario. These scenarios are generally not compatible with a different baseline scenario.

Alternative scenarios are generated or loaded from the Scenarios tab in the control panel. These scenarios share the same parameters as the baseline scenario, with the exception of adjustment factors for the size of risk groups, which can be varied to represent counterfactual developments (e.g., due to interventions). You can run multiple scenario simulations in parallel, which is efficient for reducing wait times.

The results are displayed in three separate tabs in the main panel:

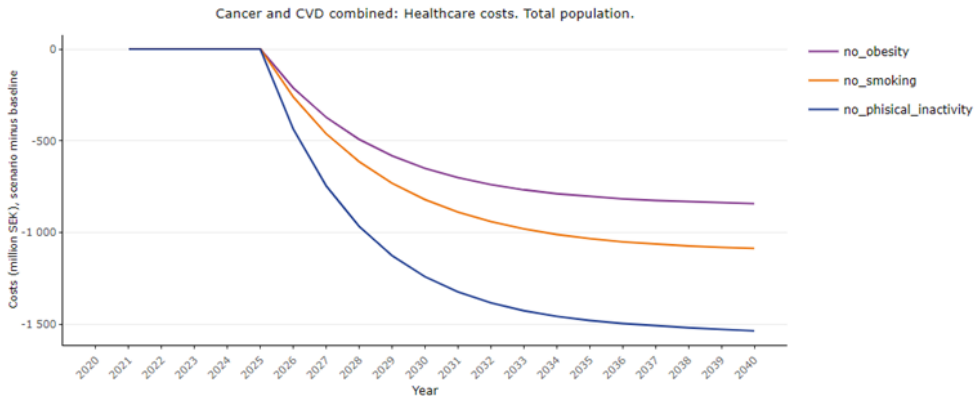
- Time series,
- Cross-sections (age distribution),
- Cumulative sums over multiple years, within scenarios and groups.

For each type of graph, you can use the drop-down menus at the top to select:

- Variable,
- Group,
- Measure.

Results can be broken down by sex or age group, either in 10-year age groups or above/below 70 years.

Figure 5 – example of a graph in NCDSim's graphical user interface



The interface is useful for comparing different scenarios with each other. In Figure 5, the development of healthcare costs in three different scenarios is shown, presented as the difference compared to the baseline scenario.

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